

ECON 301—ECONOMETRICS I

Spring 2024

Department of Economics
Nazarbayev University

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Lecture time and classroom TBA

Office hours: By appointment, book at <https://alejandromelo.youcanbook.me>, please sign at least one day in advance.

Course Description: The purpose of this course is to help students understand how to interpret economic data. To that end, you will learn how to do basic empirical research. To do empirical research means using statistical techniques to manage and analyze data. The main goal is to quantify how one economic variable (such as someone's education) affects another variable (for instance his wage). These techniques are used by both academic economists (professors) and professional economists (i.e. those working for business, government and NGOs). Academic economists use them, in large part, to test theories and models; professional economists do empirical work to help management with decision making. You will gain practical experience by solving the problems sets where you will be analyzing actual economic data.

Course Requirements: ECON 211—Economic Statistics, OR MATH 321—Probability and either MATH 322—Mathematical Statistics or MATH 310—Applied Statistical Methods.

If you have a working knowledge of the following terms, you are on good foot:

- Population, Sample, Data
- Random Variable, Probability Distribution
- Correlation, Independence
- Moments of a Distribution: Expected Value, Variance, Covariance, Correlation
- Point Estimator, Interval Estimator (Confidence Interval)
- Sampling Distribution of an Estimator
- Properties of Estimators: Unbiasedness, Consistency, Efficiency

It is your responsibility to know the prerequisites! If you need to review this material, please review your notes of ECON 211/MATH 321-322 or the appendix of Wooldridge (see below for the reference).

Course Aims:

1. Using statistical techniques to manage and analyze data.
2. Testing economic theories and models.
3. Understanding how empirical work can help policy makers in their decision making process.
4. Apply econometric tools to produce original research in applied economics.

Course Goals: At the end of the course the student will be able to:

1. Evaluate and interpret data using statistical and econometric methods.
2. Explain the conduct, results, and limitations of basic econometrics (e.g., hypothesis testing, interpreting ordinary least squares estimates, omitted variables, etc.).
3. Explain how economists use the scientific process to empirically isolate the effects of policies.
4. Apply econometric tools to produce original research in applied economics.

Learning and Teaching Methods: Lectures, in-class activities, tutorials, and problem sets.

Textbook: There are two textbooks for this course.

1. *Introductory Econometrics: A Modern Approach* by Jeffrey Wooldridge, Cengage Learning, 2020, 7th Edition. I asked the library to provide the latest edition for you in e-book format. However, any edition will suffice. There are sufficient copies of older editions in the library. If you prefer to have a physical copy of the book, please arrange with the library to have it delivered to you.
2. *A First Course in Econometrics* by Arthur Goldberger, Harvard University Press, 1991. This is a classic textbook in Econometric Theory. However, it is difficult to obtain and the library couldn't obtain enough copies. There are a few copies on the library which I have put on reserve. However, I may be able to provide you with some chapters in pdf when needed.

You will be responsible for all the material I cover or assign in class or tutorials. Therefore, studying the textbook is important but it *it is not* a perfect substitute for the material I cover in class.

Course Website: Additional course material, such as lecture slides, problem sets, and announcements will be posted on the Moodle page of the course, which you should check regularly.

Computational Requirements R with RStudio are the recommended software packages. They are free to use. You are expected to familiarize yourself with this package without losing your focus on the main material in the course. If there is another software package you are already familiar with or more interested in learning (EViews, STATA, SAS, MATLAB, etc.), you might use that package for your homework assignments. Assignments as well as exams will include R output.

Assessments: I will grade the course based on the following items:

Components	Weight
Participation	5%
Problem Sets	10%
Midterm	30%
Final Exam	35%
Research Project (more on this in the next section)	20%

Participation will be assessed via pop-up quizzes and participation during class.
In this class there are two type of assignments:

- **Homework:** This item refers to problems or questions that I will ask during the lectures that you must solve on your own. It consists of extensions of the material or filling the gaps in any of the arguments studied during the lecture. This component is ungraded and only for your own benefit, you do not need to submit it. However, anything that I ask as part of the homework can be assessed in the exams.

- Problem Sets: There will be five problem sets. You need to submit these. Late submissions will **not** be accepted. I will grade your problem sets on a scale from 0 to 2 where:
 - 0: Not delivered.
 - 1: Delivered but incomplete.
 - 2: Delivered and complete.

I will post solutions to the problem sets immediately after the deadline. *It is your responsibility to check your solutions against my solutions to see if you were correct or where you made mistakes.* Problem sets are difficult and are meant to represent the level of difficulty that you will face in the exam problems.

The midterm exam will take place **on Week 6**. I will discuss with you exactly what day works for everyone. The final exam will be scheduled by the Registrar.

I will compute your final numeric grade by computing the weighted average of the aforementioned components according to the weights given in the table above. The numeric grade will then be curved to obtain the final letter grade.

Students are expected to take all exams at the designated time. Make-up exams and homeworks will **not** be given. If you miss an exam or problem set due to serious illness, or injury of yourself or a close family member, its weight will be shifted to the remaining exams (problem sets) equally.

Research Project Finding a topic is an important part of the exercise: I won't offer suggestions (although I will be happy to respond to preliminary ideas that you come up with). The paper should be short: 5–10 pages (excluding title and reference pages) discussing the econometric analysis of a group of economic or socioeconomic variables of your choice. It should be well written. You are encouraged to use the Writing Center for help with this.

You are encouraged to work together for your research project **with a limit of maximum 5 people** per project. The project is due by **the last day of classes**. The research project grade is determined using the following rubric:

Components	Weight
Introduction and motives for the research	15%
Integration of the literature review into the paper	15%
Model and pretests	20%
Results and inference	20%
Conclusions	15%
Grammar, writing quality	15%

Grading: Your course grade is determined by the weighted average of the assessments above according to the following scale:

A	95–100	A–	90–94.99
B+	85–89.99	B	80–84.99
B–	75–79.99	C+	70–74.99
C	65–69.99	C–	60–64.99
D+	55–59.99	D	50–54.99
F	0–44.99		

Course Expectations:

1. Class will begin promptly at the scheduled starting time. As a courtesy to others, it is requested that you not arrive after class has begun or leave before it ends. PLEASE NOTE: Since showing up late and leaving early is disruptive to the class, students who disrupt the class in this or any other manner may be administratively dropped or have their course grade reduced at the instructor's discretion. If you must leave class early or will be late, please let me know before class begins.
2. Class attendance will be taken regularly throughout the semester. If you fail to attend a lecture, you must submit the documents justifying your absence to the registrar and notify me by email. Students who have more than 25% of unexcused absences (7 lectures) will not be able to take the final exam. The instructor holds the right to excuse students from class or administratively drop them if necessary for violation of this policy.

Academic Misconduct: Students are expected to act with integrity. Cheating and plagiarism will not be tolerated. Any student deemed academically dishonest will be reported to the SSH disciplinary committee. Any cases of academic misconduct will be handled in accordance with the procedures of the Student Code of Conduct. The penalties may include anything from failing grade on a particular assignment to a loss of stipend, and even, dismissal from the University. You can read more about academic misconduct in the Student Code of Conduct. https://aao.nu.edu.kz/academic_integrity

Tentative Course Outline:

Session	Topics and Assignments	Course Aims	CLOs
Week 1	Introduction: What is Econometrics? Functional Relationships in Econometrics	1	1
Week 2	Functional Relationships in Econometrics. Simple Linear Regression	1, 2, 3	1, 2, 3
Week 3	Multiple Linear Regression	1, 2	1, 2
Week 4	Multiple Linear Regression: Inference	1,2	1, 2
Week 5	Multiple Linear Regression: Inference	1, 2	1, 2
Week 6	Review & Midterm	1, 2,	1, 2
Week 7	Large Sample Theory	1,2	1, 2
Week 8	Inference and Asymptotics	1, 2	1, 2
Week 9	Further Issues	1, 2	1, 2
Week 10	Binary (or Dummy) explanatory variables	1, 2	1, 2
Week 11	Spring Break		
Week 12	More on Specification and issues	1,2,3	1,2,3
Weeks 13-15	Advanced topics: dummy dependent variables, instrumental variables, serial correlation and time series methods (depending on time and student interests).	1,2,3	1,2,3

Disclaimer: Subsequent changes can be made to any part of this syllabus when necessary. Any changes will be announced in class and on Moodle.